



EE115 Analog Circuits

模拟电路

Dr. Hao Ren 任豪

Office: SIST Bldg 3-328

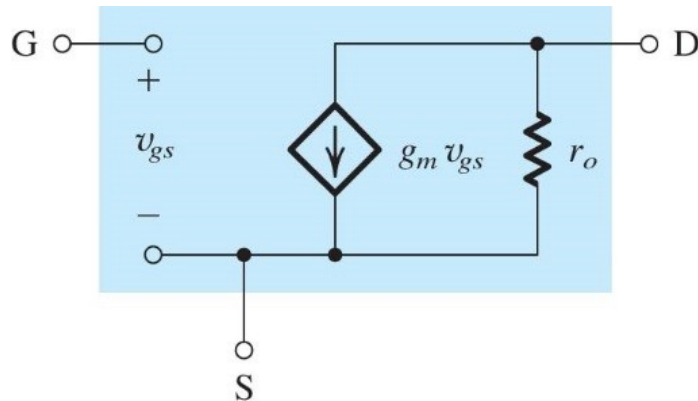
renhao@shanghaitech.edu.cn

Outline



- Transistor amplifier 2
- SEDTRA/SMITH book page 423-452;

Review-Small signal model for MOSFET



(b)

Figure 7.13 Small-signal models for the MOSFET: (b) including the effect of channel-length modulation, modeled by output resistance $r_o = |V_A|/I_D$. These models apply equally well for both NMOS and PMOS transistors.

- The equivalent circuit is valid for both NMOS and PMOS
- In PMOS, use absolute sign for all parameters: $|V_{GS}|$, $|V_t|$, $|V_{OV}|$, $|V_A|$, and replace k_n with k_p

Transconductance:

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS}=V_{GS}} \quad g_m \equiv \frac{i_d}{v_{gs}} = k_n (V_{GS} - V_t)$$

$$g_m = k_n V_{OV}$$

$$V_{OV} = \sqrt{2I_D / (k'_n (W/L))}$$

$$g_m = \sqrt{2k'_n} \sqrt{W/L} \sqrt{I_D}$$

Output resistance at drain:

$$r_o = \frac{|V_A|}{I_D}$$

$$i_D = \frac{1}{2} k_n (V_{GS} + v_{gs} - V_t)^2$$

$$= \frac{1}{2} k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t) v_{gs} + \frac{1}{2} k_n v_{gs}^2$$

$$i_d = k_n (V_{GS} - V_t) v_{gs}$$

Basic Single-Transistor Amplifier Configurations

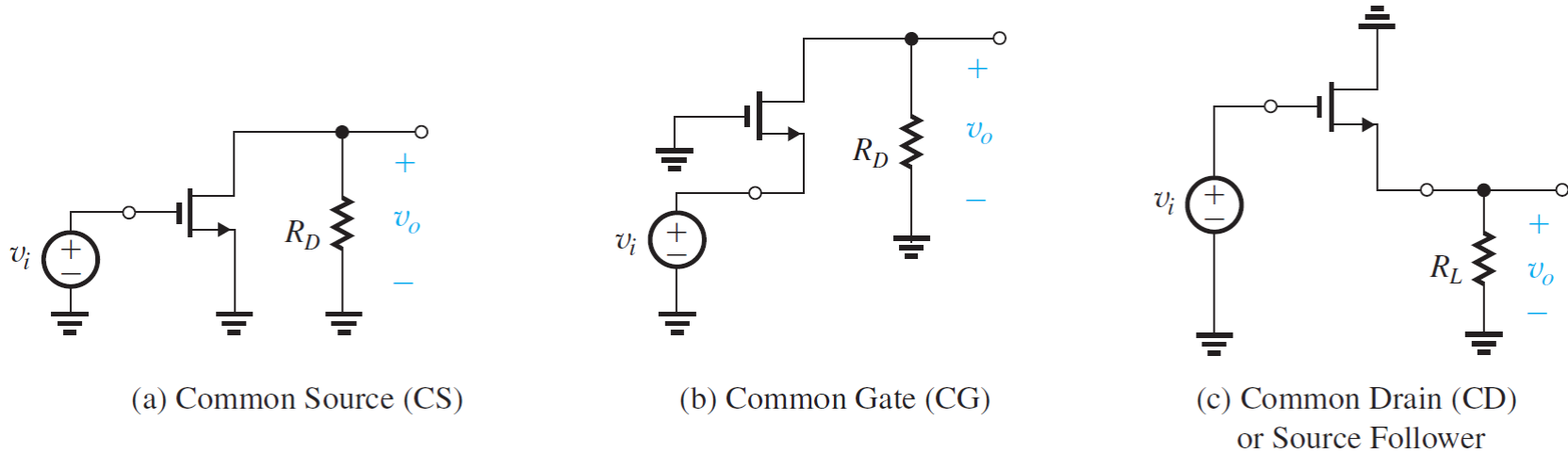
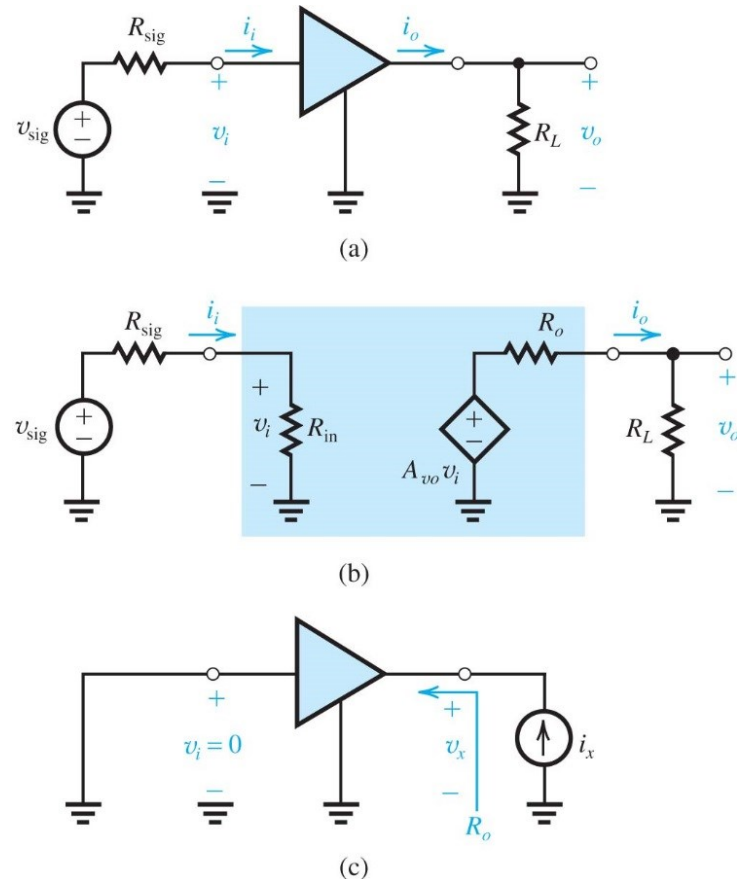


Figure 7.33 The basic configurations of transistor amplifiers. (a)–(c) For the MOSFET

Two-Port Model of Amplifiers



$$G_v = \frac{R_{in}}{R_{sig} + R_{in}} A_{vo} \frac{R_L}{R_o + R_L}$$

- In addition to **gain**, it's important to have proper **input and output resistances**.
- Procedure to find R_o :
 - Ground input, remove R_{sig}
 - Apply a test current source at output (conceptually, not experimentally),
 - Find voltage at output terminal.

$$R_o = \frac{v_x}{i_x}$$

Figure 7.34 Characterization of the amplifier as a functional block: (a) An amplifier fed with a voltage signal v_{sig} having a source resistance R_{sig} , and feeding a load resistance R_L ; (b) equivalent-circuit representation of the circuit in (a); (c) determining the amplifier output resistance R_o .

Common-Source (CS) Amplifier

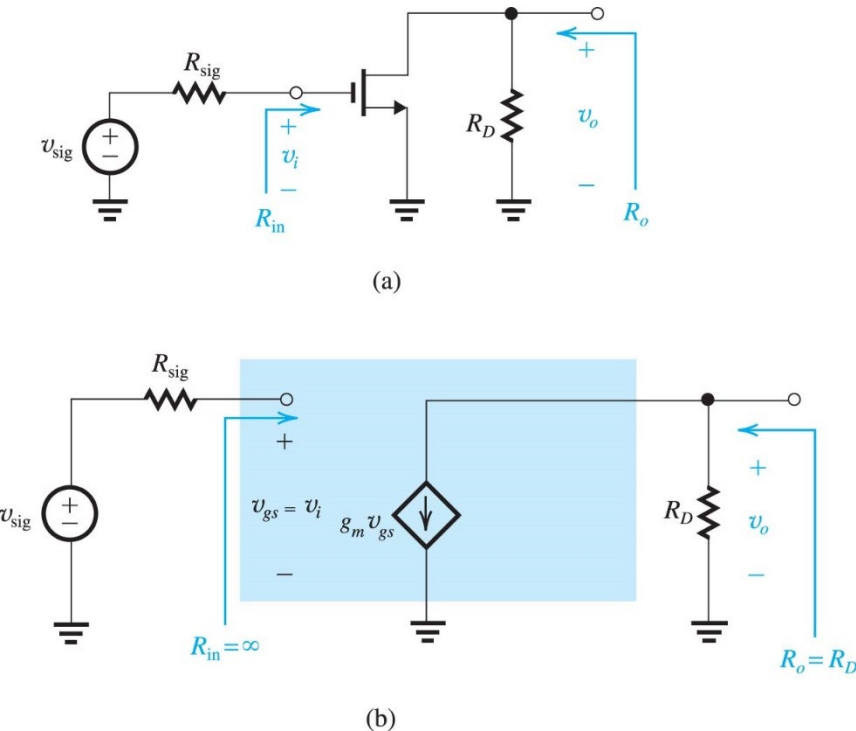


Figure 7.35 (a) Common-source amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig} . The bias circuit is omitted. (b) The common-source amplifier with the MOSFET replaced with its hybrid- π model.

- Previously, using MOSFET equation, we derived analytically :

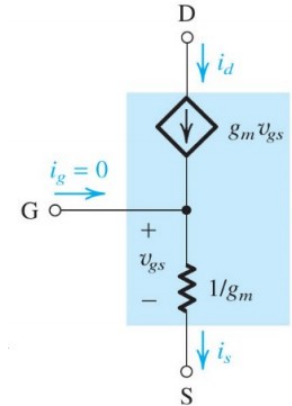
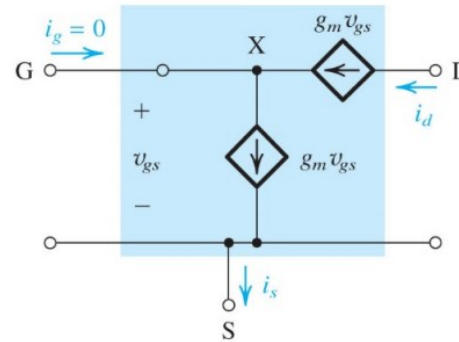
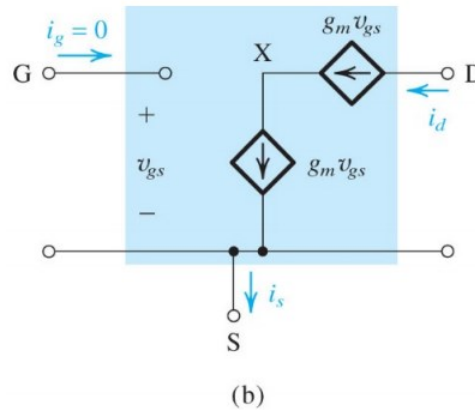
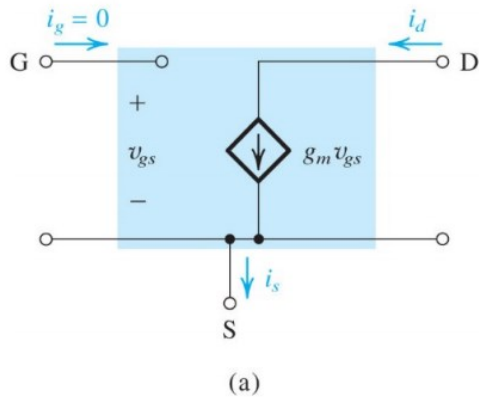
$$A_V = -k_n V_{OV} R_D$$
(The process is cumbersome, tedious, and will go quickly out of hand with more complex circuits)
- Using hybrid-pi model, it is almost trivial to solve: $A_V = -g_m R_D$
- With load resistance, R_L :
Since R_L is in parallel with R_D

$$G_V = -g_m (R_D || R_L)$$

$$R_{in} = \infty$$

$$R_o = R_D$$

T-Equivalent Circuit Model for MOSFET (without r_o)

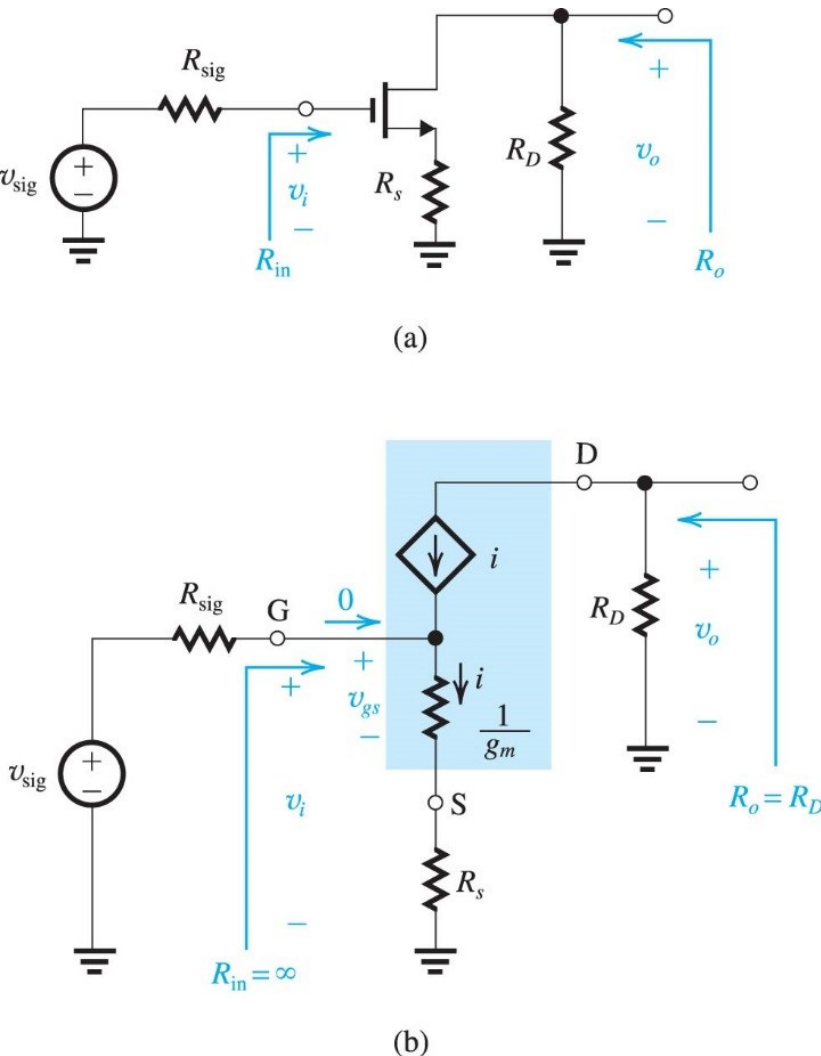


No change in circuit since the additional dependent current source has the same current.

No change in circuit since no current will flow through the new connected path

Same current with the dependent current source replaced by equivalent resistance

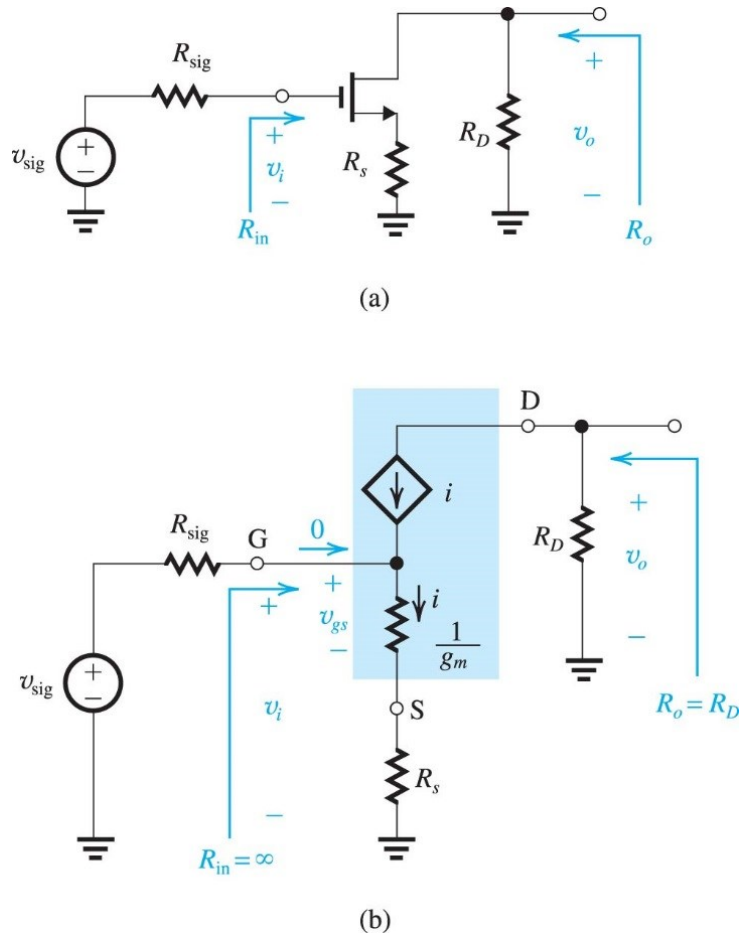
CS with Source Resistance (Source Degeneration)



- You can analyze the circuit using hybrid-pi model. (Try it !)
- However, whenever **there is a resistor connected to source**, it is much easier to use the "T-model"

Figure 7.37 The CS amplifier with a source resistance R_s : (a) circuit without bias details; (b) equivalent circuit with the MOSFET represented by its T model.

Continued



• Note

$$A_v = -\frac{R_D}{\frac{1}{g_m} + R_s} = -\frac{\text{Total resistance in Drain}}{\text{Total resistance in Source}}$$

• R_s provides negative feedback, which

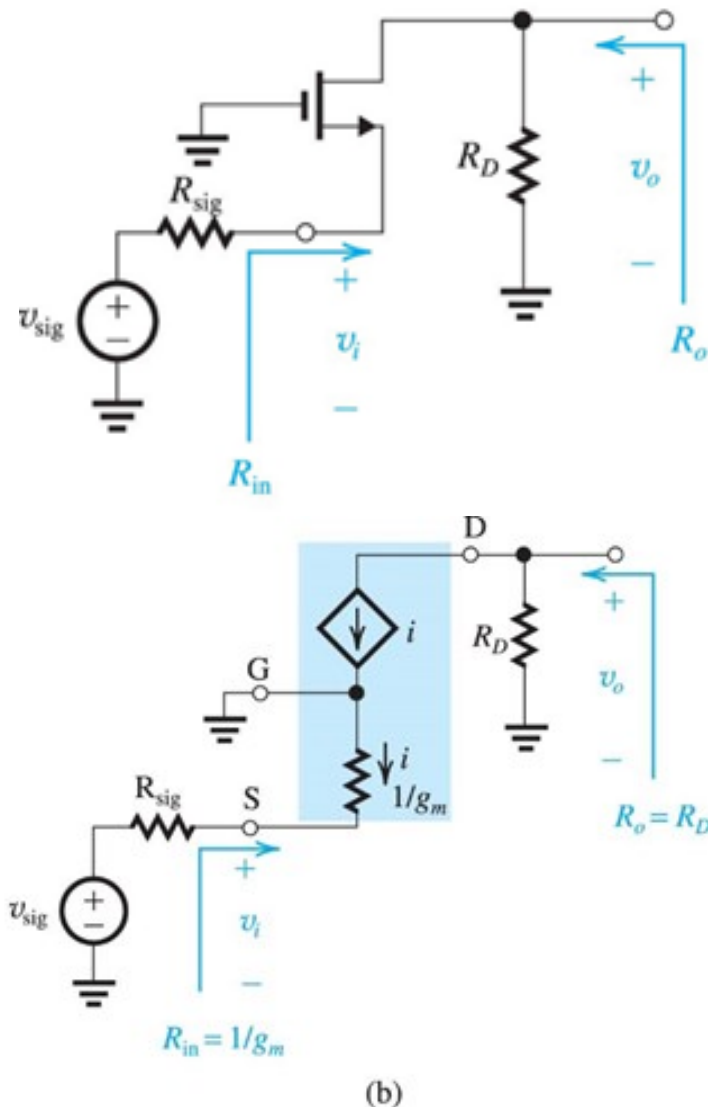
- Stabilize drain current
- Increase linearity by keeping v_{gs} small

$$v_{gs} = v_i \frac{1/g_m}{1/g_m + R_s} = v_i \frac{g_m}{1 + g_m R_s}$$

- Increase usable bandwidth

Figure 7.37 The CS amplifier with a source resistance R_s : **(a)** circuit without bias details; **(b)** equivalent circuit with the MOSFET represented by its T model.

Common-Gate (CG) Amplifier



R_{sig} connected to source
→ Use the "T-model"

$$R_{in} = \frac{1}{g_m}$$

$$A_{vo} \equiv \frac{v_o}{v_i} = g_m R_D$$

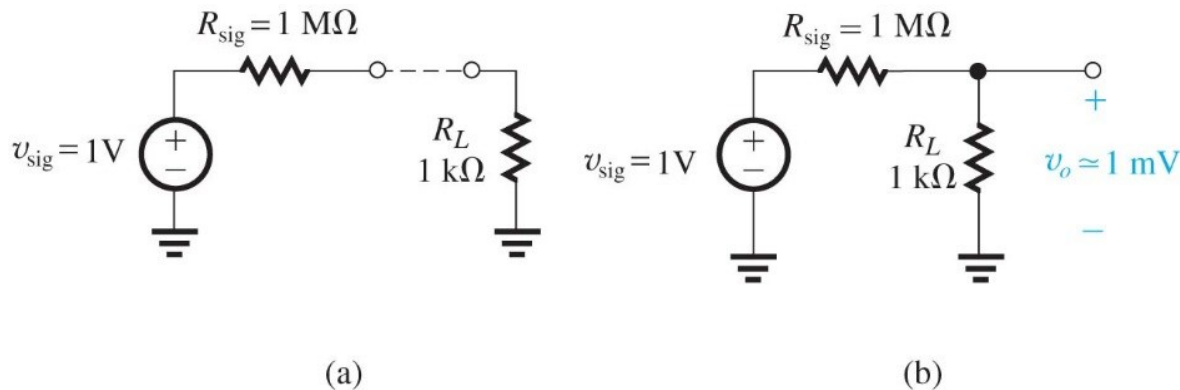
$$R_o = R_D$$

$$G_v = \frac{1/g_m}{R_{sig} + 1/g_m} [g_m (R_D \parallel R_L)]$$

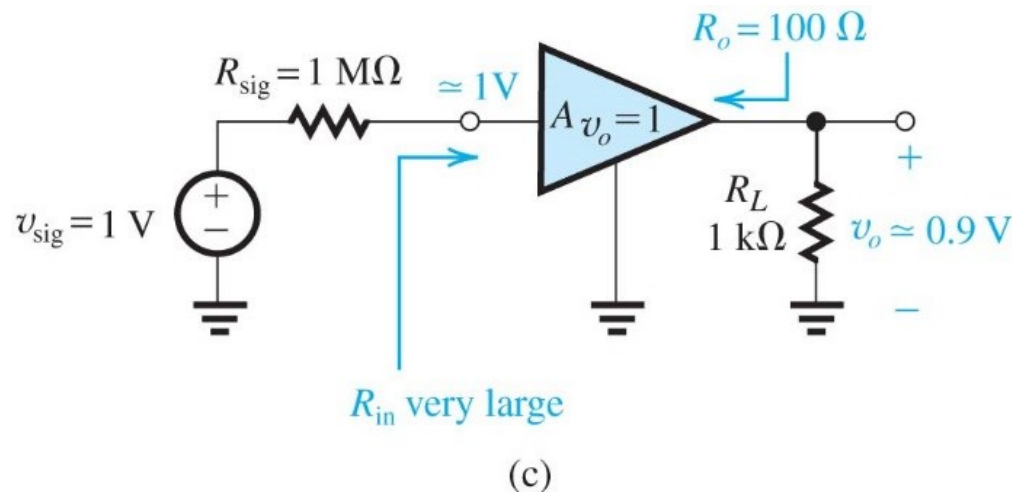
Figure 7.39 (a) Common-gate (CG) amplifier with bias arrangement omitted. (b) Equivalent circuit of the CG amplifier with the MOSFET replaced with its T model.

Unity-Gain Buffer Voltage Amplifier

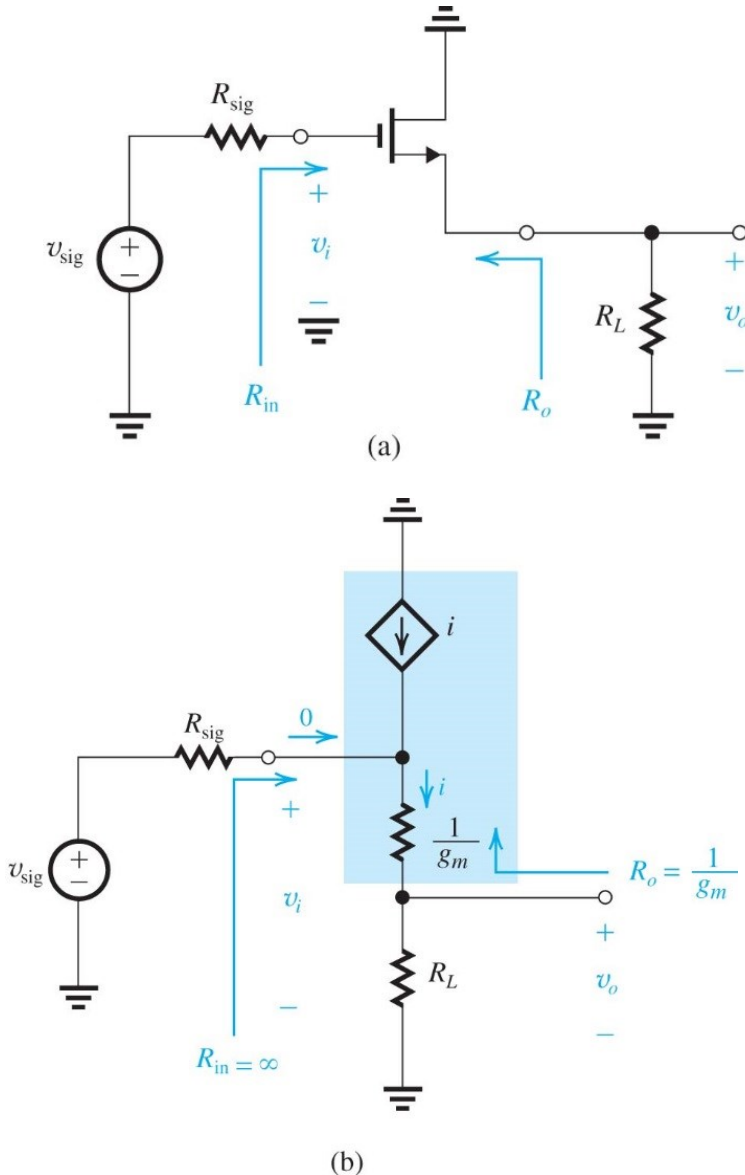
Driving low impedance load directly



Driving low impedance load with unit-gain buffer



Common-Drain (CD) Amplifier (Source Follower)



R_L connected to source
→ Use the "T-model"

$$R_{in} = \infty$$

$$A_v \equiv \frac{v_o}{v_i} = \frac{R_L}{R_L + 1/g_m}$$

$$R_o = 1/g_m$$

$$G_v = A_v = \frac{R_L}{R_L + 1/g_m}$$

Figure 7.42 (a) Common-drain amplifier or source follower with the bias circuit omitted. (b) Equivalent circuit of the source follower obtained by replacing the MOSFET with its T model.

Comparison of Different Amplifier Configurations



- CS provides the bulk of the gain
- CD used as voltage buffer in output stage



Summary of MOSFET Amplifiers

Table 6.4 Characteristics of MOSFET Amplifiers

Amplifier type	Characteristics ^a				
	R_{in}	A_{vo}	R_o	A_v	G_v
Common source (Fig. 6.35)	∞	$-g_m R_D$	R_D	$-g_m (R_D \parallel R_L)$	$-g_m (R_D \parallel R_L)$
Common source with R_s (Fig. 6.37)	∞	$-\frac{g_m R_D}{1 + g_m R_s}$	R_D	$\frac{-g_m (R_D \parallel R_L)}{1 + g_m R_s}$ $-\frac{R_D \parallel R_L}{1/g_m + R_s}$	$-\frac{g_m (R_D \parallel R_L)}{1 + g_m R_s}$ $-\frac{R_D \parallel R_L}{1/g_m + R_s}$
Common gate (Fig. 6.39)	$\frac{1}{g_m}$	$g_m R_D$	R_D	$g_m (R_D \parallel R_L)$	$\frac{R_D \parallel R_L}{R_{sig} + 1/g_m}$
Source follower (Fig. 6.42)	∞	1	$\frac{1}{g_m}$	$\frac{R_L}{R_L + 1/g_m}$	$\frac{R_L}{R_L + 1/g_m}$

^a For the interpretation of R_{in} , A_{vo} , and R_o , refer to Fig. 6.34(b).

Homework4-due April 15th 11p.m.



SEDRA/SMITH Book chapter 7 problem 7.2, 7.6,
7.25, 7.29, 7.65, 7.77, 7.82, 7.83